BitSemCom + GRAND

Results so far: does a channel code decoded with GRAND help a semantic image system?

Safaa Salman

Thesis working document, November 2026, branch `grand-implementation`

The answer in one paragraph

Does an error-correcting code decoded with GRAND improve the image quality of a bit-level semantic communication system? We measured it in two rounds on the BitSemCom models: first with the existing checkpoints for $C = 16$, 32 and 64 bits per token, then with a properly retrained $C = 32$ model and a model trained with GRAND inside the training loop, on AWGN and fading channels, with and without channel-state errors, with four quality metrics. The answer is no: the CRC(128,112) code plus ORBGRAND adds at most $+0.08$ dB PSNR on AWGN, the gains seen in fading come from the interleaver and not from the code, and the model trained with GRAND in the loop is 0.1 to 0.24 dB worse than the benchmark. The reason is that the semantic decoder already absorbs scattered bit errors, so there is almost nothing left for a code to repair. The one measurable win is for receivers that can only deliver hard bits. This document explains the experiments and the numbers in plain language; the repository guide explains the code.

Contents

1 The question and the short answer **2**

2 What was measured and how **2**

3 Round 1: existing checkpoints, $C = 16$, 32, 64 **3**

3.1 AWGN . . . 3

3.2 Why the gain is so small: diagnostics . . . 3

3.3 Block fading . . . 4

4 Round 2: a proper benchmark and GRAND in the loop ($C = 32$) **4**

4.1 Benchmark retraining . . . 4

4.2 Training with GRAND in the loop . . . 4

4.3 The code at the receiver: AWGN . . . 6

4.4 The code at the receiver: block fading . . . 7

4.5 Hard-decision receivers . . . 8

4.6 Imperfect channel knowledge . . . 8

4.7 Guess budget: 2000 against 100,000 . . . 9

4.8 Metrics beyond PSNR . . . 9

4.9 The LLR study . . . 10

4.10 The MATLAB reference check . . . 10

5 What the results mean **10**

6 Open items and next steps **11**

1 The question and the short answer

BitSemCom turns an image into a few thousand bits, sends them as QPSK symbols, and rebuilds the image from the received *soft values* (the probability that each bit is a 1). There is no channel code: the semantic decoder is trained on noisy inputs and copes with bit errors by itself. The thesis asks whether adding a code, decoded with GRAND, helps.

We added a CRC(128,112) code (16 parity bits per 112 data bits) with an interleaver between the bit mapper and the decoder, decoded it with ORBGRAND, and compared image quality with and without it. When ORBGRAND cannot decode a block within its guess budget it gives up ("abandons") and the block is passed on as plain soft values, exactly as without the code. So the code can only help where ORBGRAND decodes, and can only hurt when it "repairs" a block to a wrong codeword (a *miscorrection*).

Short answer

The code plus GRAND does not improve image quality in any setting that matters:

- AWGN: at most $+0.08$ dB PSNR (at 6 dB SNR); nothing or a small loss below 6 dB. The code also spends 16% more channel uses, worth about 0.2 dB if spent on more semantic bits instead.
- Fading: the gains that look like GRAND's ($+0.06$ to $+0.26$ dB) come from the interleaver. The same interleaver without the code does as well or better below 10 dB.
- Training with GRAND in the loop: the resulting model is 0.09 to 0.24 dB worse than the benchmark with every receiver, including with perfect bits. It is simply a worse autoencoder.
- The one real win: receivers that can only deliver hard bits (0 or 1). There the code plus GRAND gained $+0.1$ to $+0.4$ dB in round 1.

2 What was measured and how

The system. A 256×256 image becomes 256 tokens of C bits each ($C = 32$: 8192 bits). Our coded link pads the bits to blocks of 112, adds 16 parity bits per block (CRC polynomial `0x9eb2`, rate 0.875), interleaves, sends QPSK symbols through the channel, computes one LLR per bit, and decodes each block with 1-line ORBGRAND. For $C = 32$ that is 74 blocks and 4736 symbols instead of 4096. The guess budget is 2000 in the main sweep (the budget used in training) and 100,000 in a separate budget study.

Models. Round 1 used the existing stage-2 checkpoints for $C = 16$, 32, 64 (fine-tuned on a noisy channel with the original script, fixed learning rate, SNR $[-3, 6]$ dB). Round 2 added two $C = 32$ models trained with our trainer from the stage-2 checkpoint: *bench* (30k steps, warmup + cosine learning rate $10^{-4} \to 10^{-6}$, SNR drawn per batch from $[0, 10]$ dB) and *grand* (the identical recipe with CRC + ORBGRAND in the forward pass of every step, soft fallback).

Model	Training	GRAND in the loop	PSNR with perfect bits (dB)
stage2	the old checkpoint, fixed learning rate	no	23.46
bench	30k steps, warmup + cosine, SNR 0–10 dB	no	24.22
grand	identical recipe to bench	yes, budget 2000, soft fallback	24.02

Table 1: The three $C = 32$ models of round 2.

Receivers. Perfect bits (the ceiling); the plain BitSemCom soft receiver; the same with the interleaver and no code; hard 0/1 decisions; the coded receiver (interleaver + CRC + ORBGRAND, soft fallback). All receivers get the same bits and the same channel noise for a given image and SNR, so every comparison is a *paired* difference averaged over the 24 Kodak 256×256 crops, reported with its standard error (s.e.) and the number of images on which the first receiver wins.

Channels. AWGN with perfect channel knowledge; AWGN where the receiver's SNR estimate is 3 dB too high or too low; block Rayleigh fading with one gain per token (8 symbols); the same fading with a channel-estimate error of variance 0.1. SNR (E_s/N_0 per symbol) from 0 to 12 dB.

Metrics. PSNR is the headline number (differences under about 0.03 dB are noise). MS-SSIM, LPIPS and CLIP

similarity were computed as well (section 4.8).

3 Round 1: existing checkpoints, $C = 16$, 32, 64

3.1 AWGN

Table 2 shows PSNR for $C = 16$; $C = 32$ and 64 behave the same (see `docs/grand_results.md`). The *ceiling* is what a perfect channel would add (perfect bits minus the plain receiver); the *gain* is what the code added.

E_s/N_0 (dB)	perfect bits	plain soft	hard 0/1	coded soft	ceiling	gain
2	22.25	21.69	21.32	21.29	+0.56	−0.41
4	22.27	22.03	21.90	21.86	+0.24	−0.17
5	22.24	22.08	21.98	22.04	+0.16	−0.04
6	22.24	22.16	22.11	22.20	+0.08	+0.04
7	22.25	22.20	22.18	22.25	+0.05	+0.05
8	22.24	22.22	22.22	22.24	+0.02	+0.02
10	22.26	22.26	22.26	22.26	0.00	0.00

Table 2: Round 1, $C = 16$, AWGN, CRC(128,112), budget 100,000, PSNR in dB over 24 patches.

Reading: where ORBGRAND works (from about 6 dB) the ceiling itself is under 0.1 dB, and the code reaches it from 7 dB. Below 6 dB the code is harmful at this budget: 37 to 78% of the blocks are miscorrected into confidently wrong bits. Alternatives tried in round 1: a smaller budget caps the harm but creates no gain; a stronger random linear code (128,104) with 24 parity bits removes the miscorrections (never hurts, at most +0.07 dB) at 23% overhead; a short CRC(64,48) gives the best single number, +0.13 dB at 5 dB, at 33% overhead. At equal energy per semantic bit nothing beats the plain receiver by more than 0.07 dB.

Bandwidth. Perfect-bit PSNR is 22.25 / 23.46 / 24.44 dB for $C = 16$ / 32 / 64, about 1.1 dB per doubling of the bits. The code's 14 to 16% extra channel uses would therefore buy about 0.2 dB if spent on semantic bits. Every coding gain measured is smaller than that.

3.2 Why the gain is so small: diagnostics

Four facts about the semantic decoder

- **The decoder tolerates about 5% random bit noise.** The bit mapper is stochastic: re-sampling it flips about 5% of the bits and changes PSNR by nothing (22.23 vs 22.23 dB). The channel at 6 dB flips 2.3%.
- **Scattered errors are cheap, clustered errors are expensive.** Flipping 2% of the bits at random costs 0.1 dB; wiping out 2% of whole tokens costs 1.1 dB (table 3). Which bits are hit barely matters; concentration is what matters. A code fixes scattered errors, which the decoder does not need; an interleaver breaks up clusters, which it does need.
- **No channel of the bitstream is special.** Flipping 10% of one channel's bits costs 0.03 to 0.06 dB for every one of the 16 channels, so unequal error protection has nothing to target.
- **The SNR conditioning of the backbone is inert** in these checkpoints: telling the decoder "the bits are clean" after GRAND changes nothing.

flipped	random positions	most-confident bits	least-confident bits	whole tokens
$C = 16$, 1%	22.17	22.16	22.22	21.53
$C = 16$, 2%	22.12	22.09	22.21	21.16
$C = 16$, 5%	21.85	21.79	22.21	19.85
$C = 16$, 10%	21.36	20.84	22.17	18.01
$C = 64$, 2%	24.31	24.26	24.42	22.55
$C = 64$, 10%	23.47	22.57	24.40	17.98

Table 3: PSNR (dB) after flipping a fraction of the bits in different ways; perfect bits give 22.23 ($C = 16$) and 24.44 ($C = 64$).

3.3 Block fading

With one Rayleigh gain per token, a deep fade destroys a whole token: the plain receiver loses 0.6 dB ($C = 16$) to 1.0 dB ($C = 64$) at 6 dB average SNR. Interleaving alone, with no code, recovers most of that (+0.23 to +0.50 dB at 6 dB). Hard decisions are much worse than soft in fading (−0.6 to −1.1 dB): faded bits arrive as probability 0.5 and the decoder treats them like erasures; a hard decision throws that away. The code plus interleaver reaches the perfect-bit ceiling from about 14 dB average SNR, +0.06 to +0.07 dB above interleaving alone; at 6 to 10 dB it is below interleaving alone because of miscorrections. Coding without interleaving is worse than doing nothing at every SNR.

Hard-bit receivers. Against a receiver that can only deliver 0/1 decisions, the code with hard-decision fallback gained +0.09 dB at 6 dB AWGN and +0.36 / +0.23 dB at 6 / 10 dB in fading ($C = 16$). This is the one setting where GRAND earns its place.

4 Round 2: a proper benchmark and GRAND in the loop ($C = 32$)

4.1 Benchmark retraining

The benchmark run raised the validation score (mean PSNR over 0, 3, 6 and 10 dB) from 23.07 to 23.65 dB in 30k steps (5.8 h on an L4 GPU). On the full evaluation it beats the old checkpoint by +0.27 to +0.75 dB on AWGN and +0.14 to +0.69 dB in fading with the plain receiver, the gain growing with SNR (figure 1). The one place it loses is the hard-decision receiver at 0 dB (−0.35 dB): the old model was trained down to −3 dB.

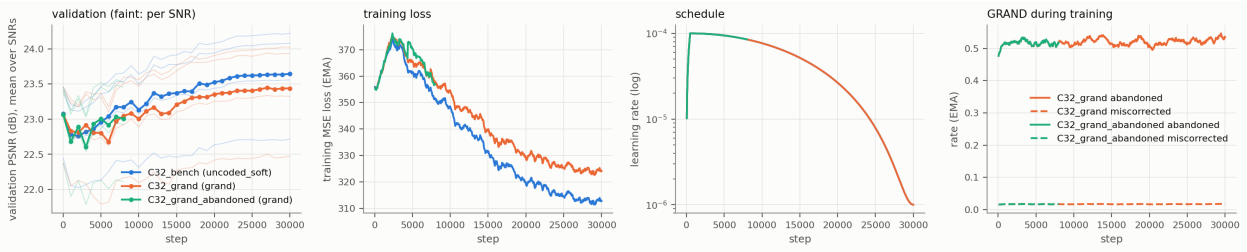


Figure 1: Training curves of both runs: validation PSNR, training loss, learning-rate schedule and the abandonment and miscorrection rates ORBGRAND reported inside the loop. Blue: benchmark. Orange: GRAND in the loop (ends 0.2 dB lower). Green: a first GRAND attempt stopped at 8000 steps by early stopping.

4.2 Training with GRAND in the loop

Same recipe, same starting weights; in every step the bits went through CRC encoding, the channel and ORBGRAND (budget 2000, soft fallback) before the decoder. The gradient to the encoder was taken from the uncoded soft channel, so both runs share one gradient path and differ only in the forward signal the decoder sees.

Run	start	best score	at step	0 dB	3 dB	6 dB	10 dB	perfect bits
bench (plain soft receiver)	23.07	23.65	30,000	22.72	23.56	24.08	24.22	24.22
grand (its own coded receiver)	23.07	23.45	27,000	22.50	23.34	23.94	24.01	24.04

Table 4: Validation PSNR (dB) on the 24 Kodak crops, AWGN; score = mean over the four SNRs.

The GRAND-trained model is worse than bench with every receiver, in every condition: on AWGN by 0.09 dB (0 dB) to 0.24 dB (4 dB) with its own coded receiver, by 0.17 to 0.22 dB with perfect bits, and by 0.16 to 0.23 dB with the plain soft receiver (s.e. 0.02 to 0.04); figure 2. The gap is the same with perfect bits as with the GRAND receiver, so this is a worse autoencoder, not a model specialised to its decoder. Its coding gain equals bench's (+0.10 vs +0.08 dB at 6 dB), so the decoder did not learn to exploit ORBGRAND's output either. Inside the loop the GRAND statistics never moved: about 50% of blocks abandoned and 1.5% miscorrected from step 50 to step 30,000, because with fixed-power QPSK the encoder has no handle on which blocks decode. The training loss stayed about 3% above bench's for the whole run, which points at the gradient surrogate.

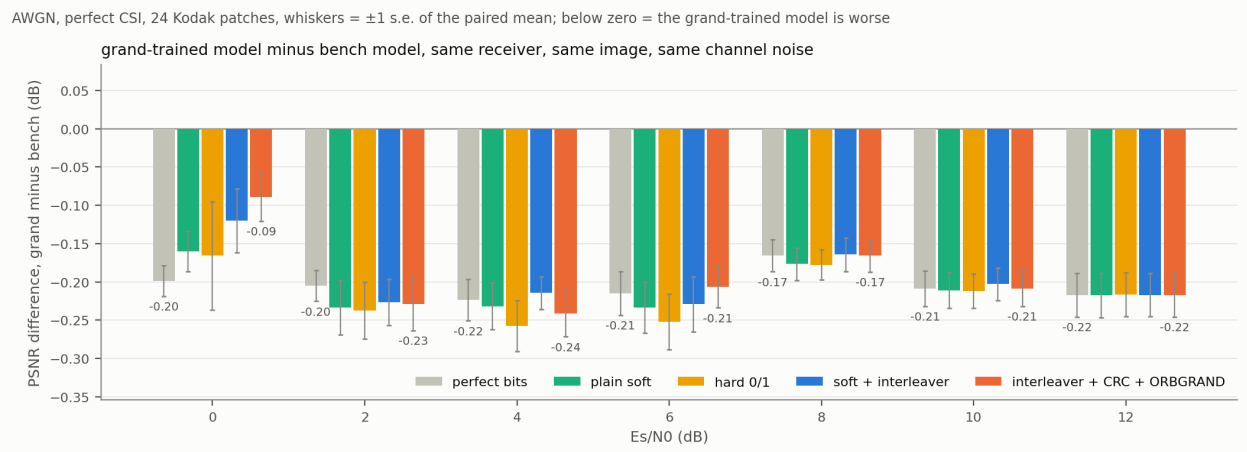


Figure 2: GRAND-trained model minus benchmark on AWGN, same receiver, same image, same noise. Every bar is below zero, including with perfect bits (grey); the coded receiver (orange) is no closer to zero than the others.

4.3 The code at the receiver: AWGN

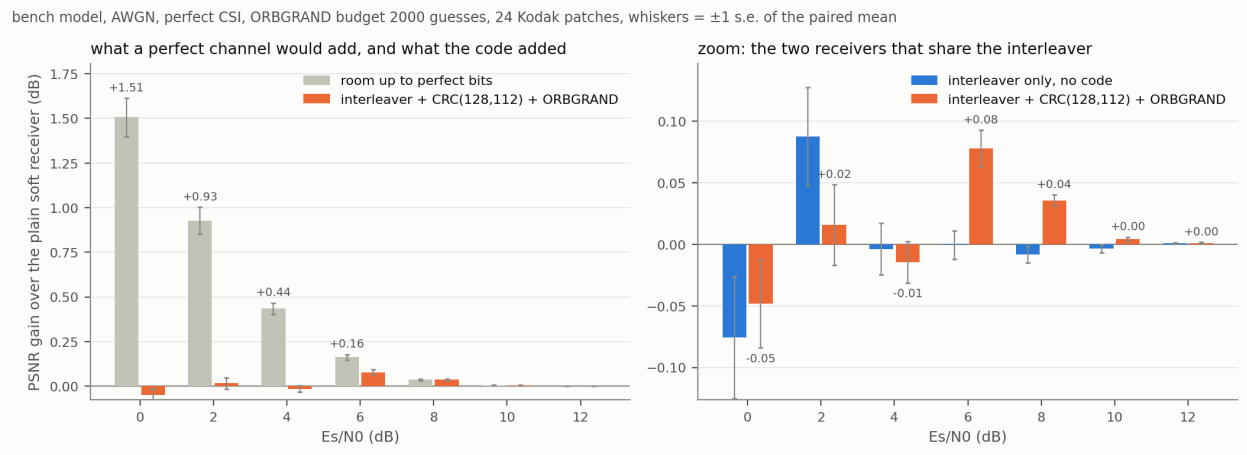


Figure 3: Bench model on AWGN. Left: how much PSNR is still available if every bit arrived correctly (grey) next to what the code plus ORBGRAND gained (orange). Right: zoom on the two receivers that share the interleaver. Whiskers are ±1 s.e. of the paired mean over 24 images.

E_s/N_0	perfect bits	plain soft	+ interleaver	+ code	code − interleaver (wins)	abandoned	miscorrected
0	24.21	22.70	22.63	22.66	$+0.03 \pm 0.05$ (15/24)	97.6%	2.4%
2	24.23	23.30	23.39	23.31	-0.07 ± 0.04 (12/24)	97.0%	3.0%
4	24.23	23.79	23.79	23.78	-0.01 ± 0.02 (11/24)	91.0%	3.0%
6	24.23	24.07	24.07	24.15	$+0.08 \pm 0.01$ (20/24)	24.7%	0.7%
8	24.22	24.18	24.17	24.22	$+0.04 \pm 0.01$ (24/24)	0.1%	0.0%
10	24.21	24.21	24.21	24.21	$+0.01 \pm 0.00$ (19/24)	0%	0%
12	24.22	24.22	24.22	24.22	0.00 ± 0.00 (13/24)	0%	0%

Table 5: Bench model, AWGN, PSNR in dB, budget 2000. “Abandoned” and “miscorrected” are fractions of the 74 blocks per image.

Where ORBGRAND can decode (6 dB and up) the room up to perfect bits is 0.16 dB or less, and the code recovers half of it at 6 dB and all of it at 8 dB. Below 6 dB the decoder gives up on 91 to 98% of the blocks and miscorrects 2 to 3%, so the coded receiver is level with the plain one or slightly below. The coded receiver wins on 20 of 24 images at 6 dB and 24 of 24 at 8 dB, but by 0.08 and 0.04 dB. Figure 4 shows that the difference is invisible.



Figure 4: One patch at 4 dB, same bits and same noise for every receiver. Top: AWGN, the four soft-input receivers are indistinguishable. Bottom: block fading, the interleaved receivers are 0.1 to 0.2 dB above the plain one and the hard-decision receiver shows the damage of whole faded tokens.

4.4 The code at the receiver: block fading

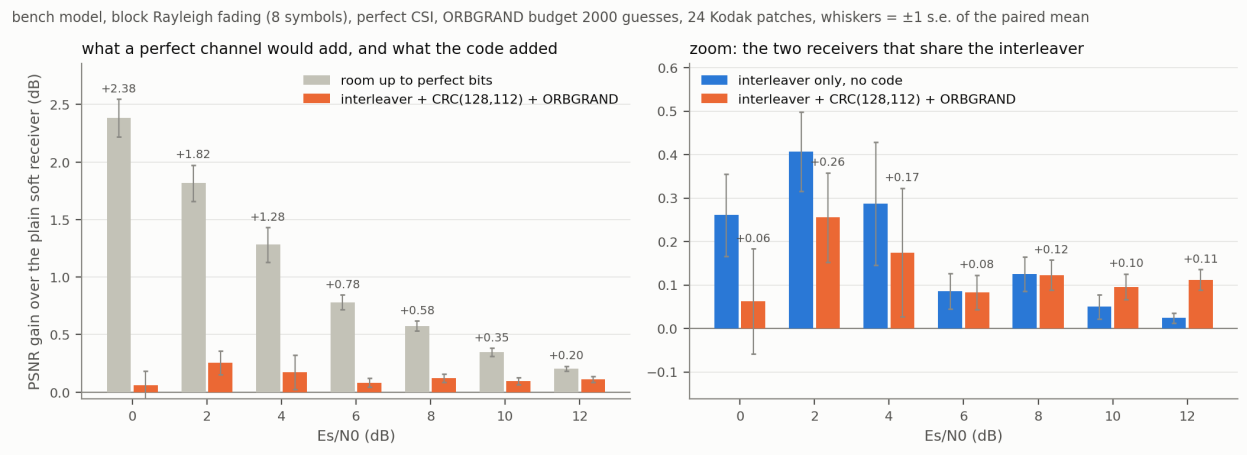


Figure 5: Bench model in block Rayleigh fading (one fade per token). Left: room up to perfect bits (grey) against the coded gain (orange). Right: the interleaver alone (blue) against the interleaver plus code (orange), both measured against the plain receiver.

E_s/N_0	perfect bits	plain soft	+ interleaver	+ code	code − interleaver (wins)	abandoned	miscorrected
0	24.21	21.83	22.09	21.89	-0.20 ± 0.07 (6/24)	96.7%	3.3%
2	24.23	22.41	22.82	22.67	-0.15 ± 0.05 (5/24)	96.4%	3.6%
4	24.23	22.95	23.23	23.12	-0.11 ± 0.06 (10/24)	97.2%	2.8%
6	24.23	23.45	23.54	23.54	0.00 ± 0.05 (11/24)	96.3%	3.4%
8	24.22	23.64	23.77	23.77	0.00 ± 0.03 (12/24)	93.5%	2.8%
10	24.21	23.87	23.92	23.96	$+0.05 \pm 0.02$ (17/24)	68.1%	2.4%
12	24.22	24.02	24.04	24.13	$+0.09 \pm 0.02$ (20/24)	31.6%	1.2%

Table 6: Bench model, block Rayleigh fading (8 symbols per fade), PSNR in dB, budget 2000.

In fading the plain receiver loses up to 2.4 dB because a deep fade wipes out whole tokens. The interleaver alone recovers +0.26 / +0.41 / +0.29 dB at 0 / 2 / 4 dB by spreading those errors over the image. The coded receiver uses the same interleaver and sits below it at 0 to 4 dB (miscorrections) and level with it at 6 to 8 dB. Only from 10 dB, where abandonment falls from 97% to 68% and then 32%, does the code add +0.05 to +0.09 dB on top of the interleaver. Against the plain receiver the coded numbers read +0.06 to +0.26 dB: that is the interleaver, not the code.

4.5 Hard-decision receivers

Receiver	AWGN 0 dB	AWGN 6 dB	fading 0 dB	fading 6 dB
plain soft	22.70	24.07	21.83	23.45
hard 0/1	21.03	23.94	19.13	22.45
interleaver + code, soft fallback	22.66	24.15	21.89	23.54

Table 7: Bench model, PSNR in dB.

A number to quote carefully

Hard decisions cost 1.7 dB at 0 dB AWGN and 2.7 dB at 0 dB fading. The soft-fallback coded receiver beats the hard receiver by the same amounts, but that measures soft versus hard input, not GRAND: at 0 dB 97% of the blocks are abandoned and arrive as soft values. The honest hard-bit number is the round-1 comparison of the code *with hard fallback* against hard decisions: +0.09 dB at 6 dB AWGN, +0.36 and +0.23 dB at 6 and 10 dB in fading.

4.6 Imperfect channel knowledge

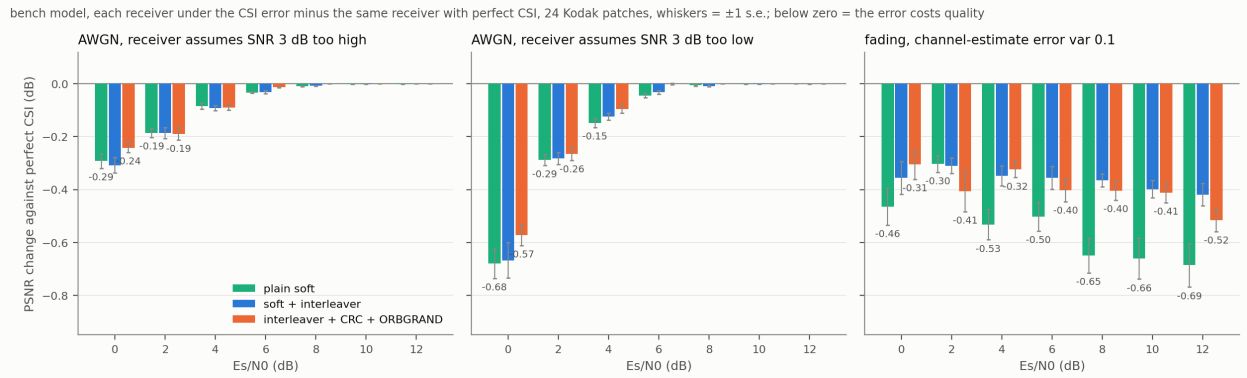


Figure 6: PSNR change of each receiver under a channel-state error, against the same receiver with perfect knowledge (bench model; below zero = the error costs quality).

An SNR over-estimate of 3 dB costs the plain receiver 0.29 dB at 0 dB and nothing from 8 dB on; an under-estimate costs 0.68 dB at 0 dB, 0.29 at 2 dB, 0.15 at 4 dB. The coded receiver pays almost the same below 6 dB (0.24 and 0.57 dB at 0 dB) because its abandoned blocks fall back to the same miscalibrated soft values, and nothing from 6 dB on. A channel-estimate error of variance 0.1 in fading costs the plain receiver 0.46 to 0.69 dB, growing with SNR, the interleaved receiver 0.31 to 0.42 dB, and the coded receiver 0.31 to 0.52 dB; with that error the block failure rate stays at 94 to 98% up to 12 dB, so the code never decodes and never beats the interleaver alone.

ORBGRAND itself is immune to an SNR misestimate: its guessing order depends only on the ranking of the LLRs, which a scale error does not change (section 4.9). This does not show in image quality because the coded receiver's quality below 6 dB is set by its abandoned blocks, which are soft values.

4.7 Guess budget: 2000 against 100,000

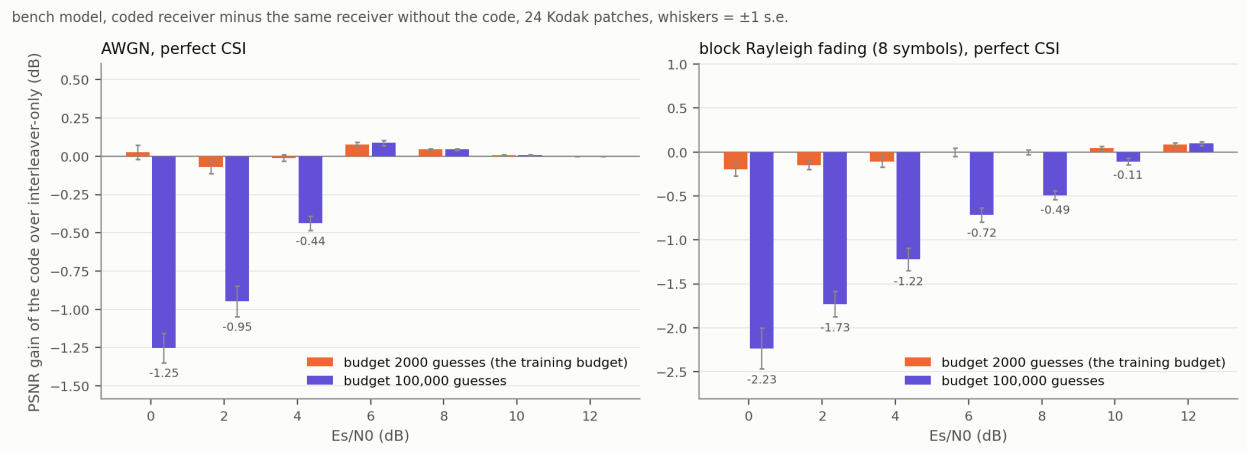


Figure 7: Gain of the code over the interleaver-only receiver at the two budgets, bench model.

A larger budget turns abandoned blocks into miscorrected ones. Below 6 dB AWGN and below 10 dB fading, 67 to 81% of the blocks come back as confidently wrong codewords and the coded receiver loses 0.4 to 1.3 dB (AWGN) and 0.5 to 2.2 dB (fading). Where the code works (6 to 8 dB AWGN, 12 dB fading) the two budgets give the same gain. With 16 parity bits the code is too weak for the error rates below 6 dB; the budget only decides whether a failed block is declared (abandonment, harmless) or hidden (miscorrection, harmful).

4.8 Metrics beyond PSNR

Condition, E_s/N_0	PSNR (dB)	MS-SSIM	LPIPS (lower is better)	CLIP similarity
AWGN 6 dB	$+0.078 \pm 0.013$	$+0.004 \pm 0.001$	-0.003 ± 0.001	-0.007 ± 0.004
AWGN 8 dB	$+0.044 \pm 0.006$	$+0.002 \pm 0.000$	-0.002 ± 0.001	-0.001 ± 0.001
fading 0 dB	-0.198 ± 0.074	-0.005 ± 0.003	$+0.005 \pm 0.004$	-0.005 ± 0.010
fading 2 dB	-0.152 ± 0.046	-0.009 ± 0.002	-0.002 ± 0.004	$+0.007 \pm 0.010$
fading 12 dB	$+0.088 \pm 0.020$	$+0.003 \pm 0.001$	-0.004 ± 0.001	$+0.003 \pm 0.004$

Table 8: Bench model, coded receiver minus interleaver-only receiver, paired over 24 images, ± 1 s.e.

MS-SSIM and LPIPS agree with PSNR in sign wherever PSNR moves by more than 0.03 dB; the exceptions are inside their standard errors. CLIP similarity differences sit inside a standard error of about 0.01 everywhere: 24 images are too few for it. The ordering of receivers is the same under PSNR, MS-SSIM and LPIPS, so none of the new metrics changes the conclusion. For the thesis we suggest PSNR plus LPIPS, with CLIP only on a few hundred images.

4.9 The LLR study

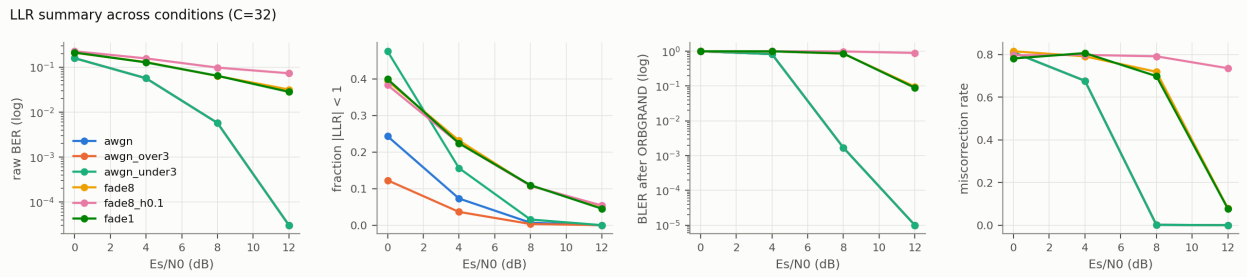


Figure 8: Raw bit error rate, fraction of weak LLRs ($|\text{LLR}| < 1$), block error rate after ORBGRAND and miscorrection rate per condition, stage2 model, large guess budget. The three AWGN curves lie on top of each other in every panel except the second.

Condition	E_s/N_0	raw BER	mean $\lvert\text{LLR}\rvert$	fraction $\lvert\text{LLR}\rvert < 1$	BLER after GRAND
AWGN, perfect CSI	0	0.159	2.3	0.24	1.00
AWGN, SNR assumed 3 dB too high	0	0.159	4.7	0.12	1.00
AWGN, SNR assumed 3 dB too low	0	0.159	1.2	0.48	1.00
AWGN, perfect CSI	8	0.0057	12.6	0.007	0.0017
AWGN, SNR assumed 3 dB too high	8	0.0057	25.2	0.003	0.0017
AWGN, SNR assumed 3 dB too low	8	0.0057	6.3	0.016	0.0017
block fading, perfect CSI	12	0.032	30.5	0.053	0.096
block fading, channel-estimate error 0.1	12	0.073	30.8	0.054	0.895

Table 9: The LLR study in numbers.

Two mechanisms. An SNR misestimate rescales the LLRs and changes the fraction of near-erasures the soft decoder sees, but leaves their ranking untouched, so ORBGRAND's decisions are identical in the three cases. A channel-estimate error in fading is different in kind: it raises the raw bit error rate from 3.2% to 7.3% at 12 dB, and 16 parity bits cannot absorb that, so block failures go from 10% to 90%. This is the mechanism behind "GRAND looks bad on bad channels": the code is too weak for the error rate, not the decoder.

4.10 The MATLAB reference check

Open item: the bit-exact check

The decoder used everywhere is our Python port of Ken Duffy's ORBGRAND. It reproduced the block error rates published with GRAND-MATLAB in round 1, and a test set of 740 real LLR blocks is exported for a bit-exact comparison with the unmodified MATLAB functions. That comparison has not been run yet: it needs a MATLAB session, and the Octave attempt failed on a MATLAB-only `cumsum` option.

5 What the results mean

The semantic decoder was trained to live with noisy bits, so a channel code has very little left to fix. Where GRAND works (from about 6 dB with 16 parity bits) the room up to perfect bits is under 0.2 dB and the code recovers it, which is all of almost nothing. Below 6 dB the code is too weak: blocks are abandoned (no change) or miscorrected (harm), and a bigger budget only converts the first into the second. Soft values act as erasures, so any receiver that keeps them keeps most of the robustness, code or no code. The 16% extra bandwidth would buy about 0.2 dB as semantic bits,

more than any coding gain measured. And training with GRAND in the loop cannot help through the encoder, which has no handle on block decodability; in our run it did not help through the decoder either.

For the thesis this is a clean negative result with a mechanism behind it: for a bit-level semantic system with a soft-input decoder, a high-rate code plus GRAND does not improve image quality on AWGN or fading channels, because the decoder already absorbs scattered bit errors and the interleaver handles clustered ones.

Three framings with evidence behind them

1. GRAND for receivers that only deliver hard bits (+0.1 to +0.4 dB in round 1), a realistic deployment constraint.
2. Interleaving first, with the code reaching the perfect-bit ceiling from about 10 to 14 dB average SNR in fading, for bit-exact delivery.
3. GRAND's immunity to SNR misestimates (its guessing order depends only on LLR ranking), which only pays off where GRAND actually decodes.

6 Open items and next steps

Next steps, cheapest first

- Run the bit-exact check against Duffy's MATLAB code (export ready; needs MATLAB).
- Fine-tune only the decoder of the benchmark with GRAND in the loop (encoder frozen): no gradient surrogate is involved, so it tests directly whether a decoder can profit from ORBGRAND's output.
- Equal-bandwidth comparison: train $C = 28$ and compare coded $C = 28$ against uncoded $C = 32$ at the same number of channel symbols (the interpolation predicts uncoded wins by about 0.15 dB).
- If coding stays in the design: a stronger code (24 parity bits removed the miscorrections in round 1) or soft-output GRAND, which hands the decoder probabilities instead of hard decisions.
- Use ORBGRAND's abandonment flag to retransmit failed blocks (HARQ); the flag is already logged.
- Evaluate CLIP on a few hundred images if it is kept as the semantic metric.

Where the numbers come from

Round 1: `results/C16_*.jsonl`, `C32_*.jsonl`, `C64_*.jsonl` and the tables `results/summary_*.md`, `bit_importance_*.md`, `channel_importance_*.md`; full write-up in `docs/grand_results.md`. Round 2: `results/eval_C32.jsonl` (budget 2000, 12,600 rows), `results/eval_C32_b1e5.jsonl` (budget 100,000), training logs in `results/train_logs/`, the LLR study in `results/figures/llr_summary.md`; full write-up in `docs/grand_review.md`. The figures in this document are in `docs/figures/grand_review/` and are redrawn by `experiments/plot_review_gains.py`, `plot_results.py`, `plot_training.py` and `plot_llr.py`.